

Week 3 · Class 1

Experimental Design in Political Science

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Experimental Design in Political Science

Today

- Designing your own experiment: what to vary, what to measure
- Vignette, conjoint, economic game, and field experiments
- Within- vs. between-subjects, and factorial designs
- What goes wrong: attention, attrition, compliance, spillover, p-hacking

Designing Your Own Experiment

Why This Is the Most Useful Thing You'll Learn

Most honors theses in this department are **experiments**. There's a reason.

To make a causal claim from observational data, you have to *find* the right variation in the world, defend a clever identification strategy, and gather a lot of data — and your result is still contestable.

An experiment lets you **manufacture** the comparison you need. With a Qualtrics account and a modest budget, you can build a clean causal test from scratch.

Today we stop *reading* experiments and start *designing* them.

What an Experiment Actually Is

Three moves:

1. **Vary something** (X) — the treatment
2. **Measure something** (Y) — the outcome
3. **Assign at random** — so the groups differ *only* by X

That's the whole machine. Everything else is logistics in service of those three moves.

Because assignment is random, any systematic difference in Y must be caused by X. We'll save the formal causal-inference machinery for next class — today the question is simply: **what is the effect of X on Y?**

Random Assignment Random Sampling

Random sampling decides who gets *into* your study.

Buys **external validity** — the ability to generalize to a population.

Random assignment decides who gets *which condition*.

Buys **internal validity** — a clean causal comparison.

You can have one without the other. An MTurk study has solid random assignment (good causal claim) but a strange sample (shaky generalization). A perfectly representative poll with no manipulation has neither.

For a thesis: **nail assignment first**. That's the causal engine. A weird sample is a footnote; a broken comparison is fatal.

Your Whole Study Is Two Decisions

Strip everything else away and an experiment is two choices:

- **What do you vary?** — the X (your manipulation)
- **What do you measure?** — the Y (your outcome)

Get these two right and the rest is plumbing. Get them wrong and no sample size or fancy statistics can rescue you.

We'll take each in turn.

What to Vary

From Idea to Treatment

You start with a concept. You need a concrete thing on a screen.

The ladder: **theory** → **construct** → **operationalization**.

Tomz & Weeks (2013) wanted to know: are people less willing to attack a **democracy** than an otherwise identical autocracy?

- Construct: the adversary's **regime type**
- Operationalization: a vignette about a country building nuclear weapons; one randomly varied line calls it a "democracy" vs. "not a democracy"
- **X** = adversary's regime type · **Y** = support for a military strike
- Finding: people are far less willing to strike a democracy — even one building nukes

The abstract construct ("threat," "democracy") had to become one concrete, swappable line of text.

Change ONE Thing (the Bundle Problem)

A good manipulation differs from the control on **exactly one** dimension. Otherwise you can't say what caused the effect.

Brader (2005) built campaign ads to evoke **fear** vs. **enthusiasm** — using ominous music *and* grainy imagery at once. If the fear ad "works," is it the music? The images? The combination?

That's the **bundle problem**: when your treatment changes several things, a significant effect doesn't tell you *which* thing mattered.

The fix: hold everything constant except the one feature you care about. If you must bundle, you can only claim *the bundle* works — not its parts. Know exactly what's inside your treatment.

The Control Condition Is a Choice

Your effect is **always relative to the control**. Change the control, change what the result means.

Pure control: nothing, or unrelated filler.

Answers: “does the treatment do *anything*?”

Active (placebo) control: a comparison matched on everything but the active ingredient.

Answers: “does *this specific feature* matter, beyond just getting a message?”

Test a get-out-the-vote mailer against *no mail* and almost anything looks effective. Test it against a *generic “please vote” mailer* and you isolate the special ingredient.

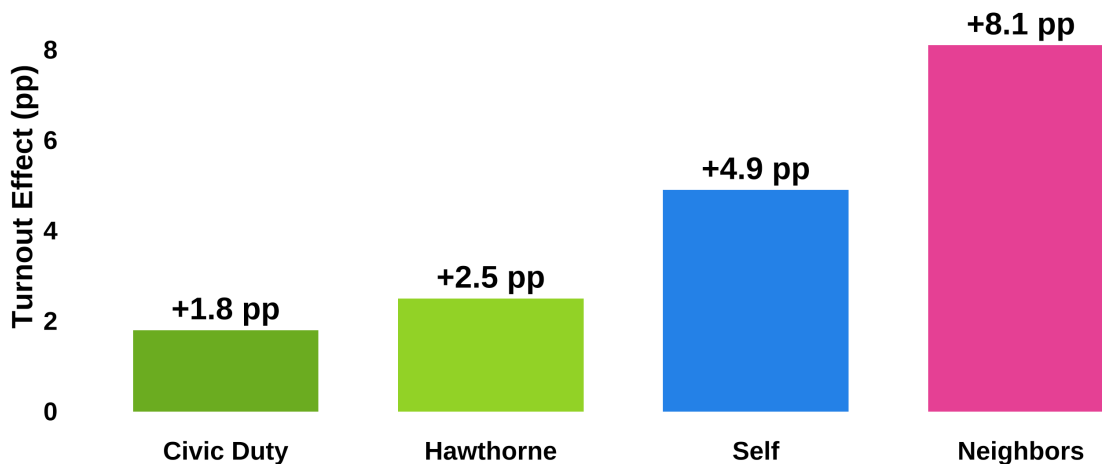
Before you run anything, ask: “**compared to what?**”

Dose: Weak Manipulations Cause False Nulls

If your treatment is too subtle, you’ll find “no effect” — and wrongly blame the *theory* when the real problem was a *weak manipulation*.

Same Message, Stronger Dose, Bigger Effect

Gerber, Green & Larimer (2008): escalating social pressure, 180,000 households



Same outcome, same population — only the **dose** changed. The strongest version rivaled door-to-door canvassing at a fraction of the cost.

Control vs. Realism

Every manipulation trades **control** against **realism**.

Vignettes give you surgical control but live in a hypothetical. Field experiments give you real behavior but you control less.

Butler & Broockman (2011) emailed ~5,000 real state legislators from a putatively Black (“DeShawn Jackson”) vs. White alias, asking for help registering to vote.

- No lab, no self-report — *actual* legislator behavior
- More realism, less control: you see *that* race moved the reply rate, not *why*

The dial:

- **High control** → vignette / survey experiment
- **Middle** → lab / economic game
- **High realism** → field experiment

Pick your spot based on your question, not on habit.

What to Measure

Behavior Beats Attitudes

Two flavors of outcome:

- **Attitudinal / self-report**: cheap and flexible (a 1–7 support scale) — but people can posture, guess, or misreport. It’s cheap talk.
- **Behavioral**: did they actually *do* it? Donate, sign, click, register, (validated) vote.

Behavioral DVs are more credible and harder to fake. Butler–Broockman measured whether the legislator *replied*; the GOTV studies use **validated turnout** from the voter file, not “did you vote?”

When you can, measure behavior — or at least an outcome with real consequences.

One Item or Many? Pick Your Primary Outcome

A **single item** is noisy. A **multi-item index** averages out noise and is more reliable — several questions combined into one “warmth” or “authoritarianism” scale.

But decide your **primary** outcome *before* you see the data. If you measure ten things and report the two that “worked,” you’re not testing a hypothesis — you’re fishing (the *garden of forking paths*).

Rule of thumb: **one** pre-specified primary DV, a short list of secondaries, and everything else labeled exploratory.

When You Measure Matters

Effects can be real *and* fleeting. Many framing and ad effects **decay within days**.

If you only measure 30 seconds after treatment, you’ve learned about an immediate blip — not whether anything *sticks*.

A delayed follow-up (a day, a week, a month later) tests **durability** — often the more interesting and more honest question for a thesis.

Decide up front: do you care about the immediate reaction, the lasting change, or both?

Demand Effects: Don’t Telegraph Your Hypothesis

If participants can guess what you’re hoping to find, some will “help” you and others will fight it. Either way you’re measuring their *read of your study*, not the real effect.

Warning signs:

- The manipulation and the outcome sit right next to each other
- The question wording makes the “expected” answer obvious
- A within-subjects design shows the same person both conditions back-to-back

Fixes: separate treatment from measurement, use neutral outcome wording, bury the key question among distractors, and don’t make the comparison transparent. The participant should not be able to reverse-engineer your hypothesis from the screen.

What Makes a Good Experimental Thesis

A checklist before you fall in love with an idea:

1. **A clear X** you can manipulate cheaply and ethically
2. **A credible Y** you can actually measure — ideally behavioral
3. **A control** that isolates the mechanism you care about (“compared to what?”)
4. **A sample** you can realistically recruit and afford
5. **Enough power** to detect an effect the size you’d expect

If any one of these is missing, fix it *now* — not after you’ve spent your budget.

Pilot Before You Commit

Run a small **pilot** before you spend real money:

- Does the manipulation actually land? (check it — see manipulation checks later)
- How long does the study take? Where do people drop out?
- Do the instructions make sense to someone who isn’t you?

A pilot of 30–50 people catches the embarrassing problems while they’re still cheap to fix.

Where to Run Your Study

Where you recruit shapes your data quality, your cost, and how far you can generalize.

- **Prolific**: clean, fast, ethical pay — but pricier
- **MTurk**: cheap and huge — but real bot and fraud risk (vet carefully)
- **CloudResearch**: pre-vetted MTurk workers — a middle ground
- **Campus subject pool**: free, but narrow (students only)
- **YouGov / Lucid**: representative samples — expensive
- **Field partner**: a campaign, school, or agency — realism, but logistics

Match the platform to the question: a tight causal test tolerates a convenience sample; a claim you need to *generalize* needs a representative one. Budget for attention checks and some attrition either way.

Break

~10 minutes — then: the menu of experimental designs

Your X and Y Take Many Forms

Everything we look at next is just a different way to **vary X** and **measure Y**:

- **Vignettes** — vary words in a scenario
- **Conjoints** — vary many attributes at once
- **Economic games** — vary incentives, measure behavior with money on the line
- **Field experiments** — vary real-world treatments, measure real behavior

Keep asking of every study: *what's the manipulation, and what's the outcome?*

Vignette Experiments

Vignettes

Present participants with a hypothetical scenario, then randomly change specific details across conditions.

Why use them?

- Isolate the effect of a single factor with tight control
- Study sensitive topics (discrimination, violence) without real-world harm
- Cheap and fast to run on survey platforms

The catch: what people say they'd do in a hypothetical isn't always what they actually do.

Example: Nuclear Weapons and Public Opinion

How strong is the norm against using nuclear weapons?

- N = ~750 (5 conditions)
- Representative U.S. sample
- YouGov/Polimetrix, January 2011
- Vignette: Al Qaeda nuclear lab in Syria
- Manipulation: nuclear vs. conventional strike effectiveness (90% vs. 90%, 70%, or 45%)
- Outcome: preference for nuclear strike
- Finding: support increases dramatically with utility — the taboo is weak

Example: Credit Claiming Messages

- N = 1,001 (MTurk)
- 2x2 factorial design
- Email-based messages

Do constituents respond more to message frequency or dollar amounts?

- Factor 1: frequency (1 vs. 5 messages)
- Factor 2: amount (\$15K vs. \$1.5M total)
- Outcome: representative evaluations
- Finding: frequency matters more than amount — 90x more effective per dollar

Grimmer et al. Treatment Text

Headline: Representative |lastName (|party, |state-|district) Brings Local Fire Departments |awardAmount for Firefighter Safety

Full text: A total of |awardAmount in grants for operations and safety programs was awarded to local fire departments from the Department of Homeland Security, Rep. |lastName announced. |firstName |lastName (|party, |state-|district) announced the grants today. Specifically, the grant will be used to improve training, equipment, and make modifications to fire stations and facilities in local fire departments. “This is great news for our local community,” said Representative |lastName. “With these funds, our local fire departments will continue to train and operate with the latest in firefighter technology.”

Key |lastName: The representative’s last name |firstName: The representative’s first name
|party: The representative’s party |state: The representative’s state |district: The representative’s district
|awardAmount: The dollar amount claimed

Conjoint Experiments

Conjoints

Vignettes manipulate one thing at a time. Conjoints manipulate many things at once.

Participants see profiles with multiple randomly varied attributes and choose between them. This reveals which attributes actually drive decisions — and which don’t matter as much as people claim.

Attribute	Immigrant 1
Prior Trips to the U.S.	Entered the U.S. once before on a tourist visa
Reason for Application	Reunite with family members already in U.S.
Country of Origin	Mexico
Language Skills	Spoke fluent English
Profession	Child care provider
Job Experience	One to two years of job training and experience
Employment Plans	Does not have a contract with a U.S. employer but has done so in the past
Education Level	Equivalent to completing two years of college in the U.S.
Gender	Female

Task: “If you had to choose between them, which of these two immigrants should be given priority to come to the United States to live?”

Example: Immigration Preferences

- N = 1,407 U.S. adults
- 9 attributes varied
- 5 profiles each
- Nationally representative

Reveals multidimensional preferences simultaneously.

Findings:

1. Economic attributes dominate (education, employment)
2. Country of origin has minimal effect — challenges the xenophobia narrative
3. Broad consensus across party lines
4. No attribute interactions detected

Vignettes vs. Conjoint

Vignettes are better when you want to isolate one factor cleanly and tell a realistic story. Conjoint are better when you want to pit many attributes against each other to see what actually drives choices.

Conjoint also reduce social desirability bias — it’s hard to guess the “right” answer when nine things are varying at once.

But conjoints can overwhelm participants, and some randomly generated profiles are implausible (e.g., a 25-year-old with a PhD and 20 years of experience).

Economic Games

Economic Games

Participants make decisions with real money on the line. The idea: if there are actual stakes, behavior in the game reflects genuine preferences rather than cheap talk.

Game	Structure	Measures	Political Application
Public Goods	Contribute to shared pool	Cooperation	Collective action
Ultimatum	Proposer offers split	Fairness norms	Distributive politics
Trust	Send money, tripled, trustee returns	Trust & reciprocity	Social capital
Dictator	Unilateral allocation	Altruism	Redistribution preferences

Example: Trust Game with Political Identity

- N = 814 participants
- Real money (\$10 stakes)
- Within-subjects design
- 2x2: Party x Race crossed

Do partisans discriminate against the other side in economic games?

1. Player 1 receives \$10 per round
2. Can allocate \$0–\$10 to Player 2
3. In trust game: amount tripled, Player 2 can return money
4. Player 2 profiles: Party (D/R) x Race (White/Black) varied

Finding: partisan discrimination exceeds racial discrimination

Trade-offs with Games

Real stakes are the big selling point — you're measuring behavior, not just attitudes.

But lab games are simple by design, and politics is not. A \$10 trust game between strangers captures something about partisanship, but it's a long way from actual political behavior. Samples also tend to skew toward students, and the stakes involved are often trivial relative to real decisions.

Field Experiments

Field Experiments

Move out of the survey and into the real world. Randomly assign treatments to actual people in natural settings and measure what they actually do.

The gold standard for external validity, but hard to pull off: expensive, logistically complex, and dependent on organizational partners.

Common in political science for studying voter mobilization, campaign effects, government service delivery, and policy interventions.

Example: Get-Out-The-Vote

- $N = 30,000$ registered voters
- New Haven, Connecticut
- 1998 election

Do personal contacts increase voter turnout?

- Control: no contact
- Treatment 1: phone calls
- Treatment 2: direct mail
- Treatment 3: face-to-face canvassing

Finding: canvassing increases turnout by 9.8 percentage points; phone calls have no effect

Example: Transparency and Corruption

- $N = 608$ villages
- East/Central Java, Indonesia
- \$3.8 million in spending
- Engineering audits

Does transparency reduce corruption?

- Control: normal oversight (4% audit probability)
- Audit treatment: 100% audit probability announced
- Invitations: more villagers invited to meetings
- Invitations + comments: invitations plus anonymous comment forms

Finding: audits reduced missing expenditures by 8.5 percentage points; community participation had little average effect

Randomization Units in Field Experiments

Who or what gets randomly assigned?

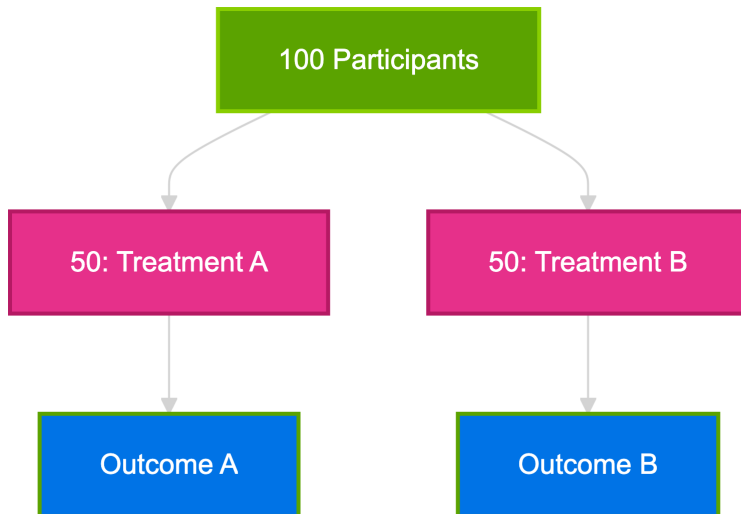
- **Individual:** voters, citizens, students
- **Cluster:** villages, precincts, schools (when individual randomization causes spillover)
- **Time:** before/after periods
- **Geographic:** districts, regions

Cluster randomization is common but reduces statistical power — you need more clusters, not just more people per cluster.

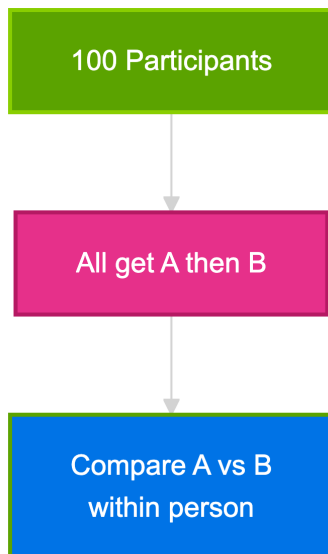
Within-Subjects vs. Between-Subjects

Two Ways to Assign Treatment

Between-subjects: each person sees one condition only.



Within-subjects: each person sees all conditions.



Within-subjects needs roughly half the sample for the same statistical power.

Clifford, Sheagley & Piston (2021)

- 12 experiments
- 27,673 participants
- Direct comparisons

Repeated measures increase precision by 35–40% without biasing treatment effects.

Within-subjects works when:

- Treatments are conceptually independent
- Order is randomized
- There's enough “washout” between conditions
- No learning effects carry over

When those conditions don't hold — when seeing one treatment changes how you respond to the next — stick with between-subjects.

Factorial Designs

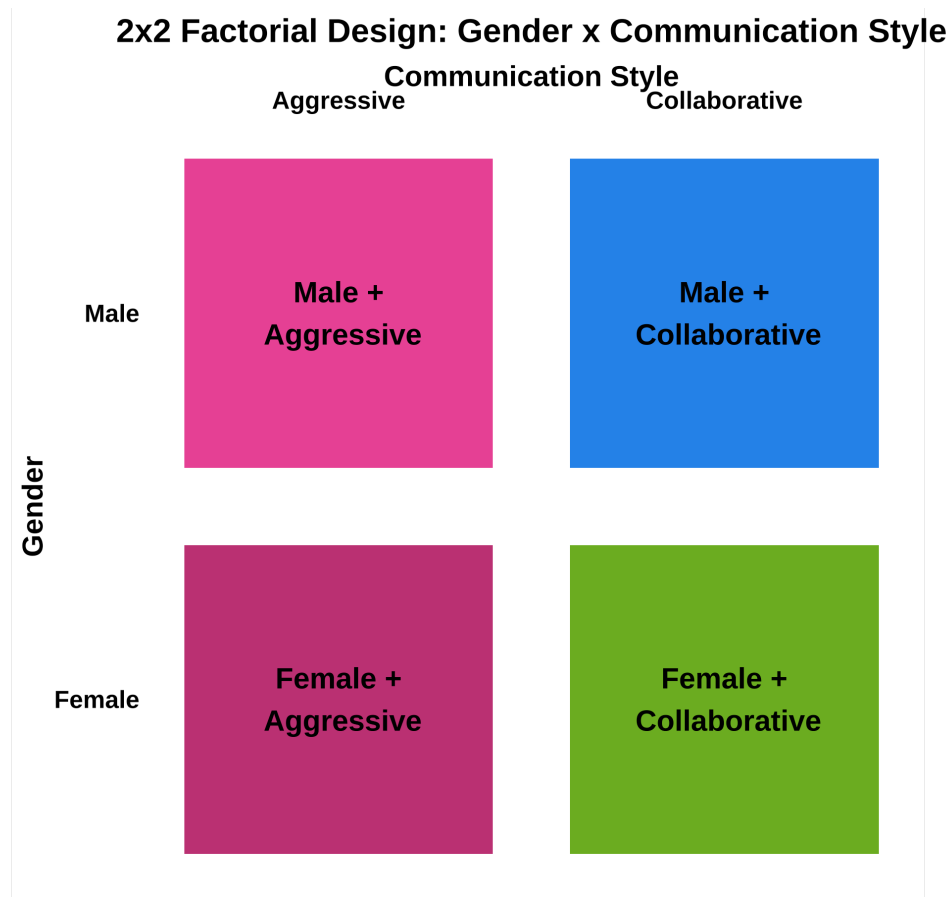
2x2 Designs

Vary two factors at once. This lets you estimate the effect of each factor separately *and* test whether their combination produces something unexpected (an interaction).

An **interaction** means the effect of one factor depends on the level of another — e.g., a resume-quality boost might help one group more than the other.

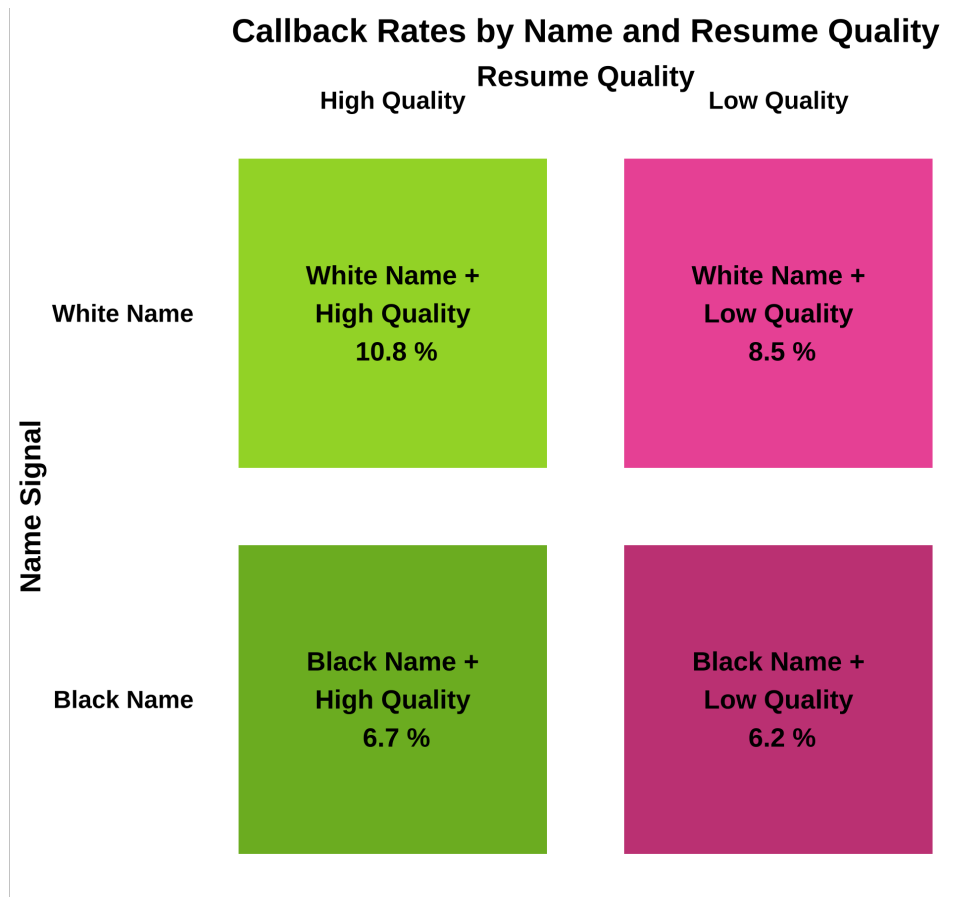
One study gives you three tests:

1. Main effect of Gender
2. Main effect of Communication Style
3. Gender x Communication Style interaction



Example: Resume Audit Study

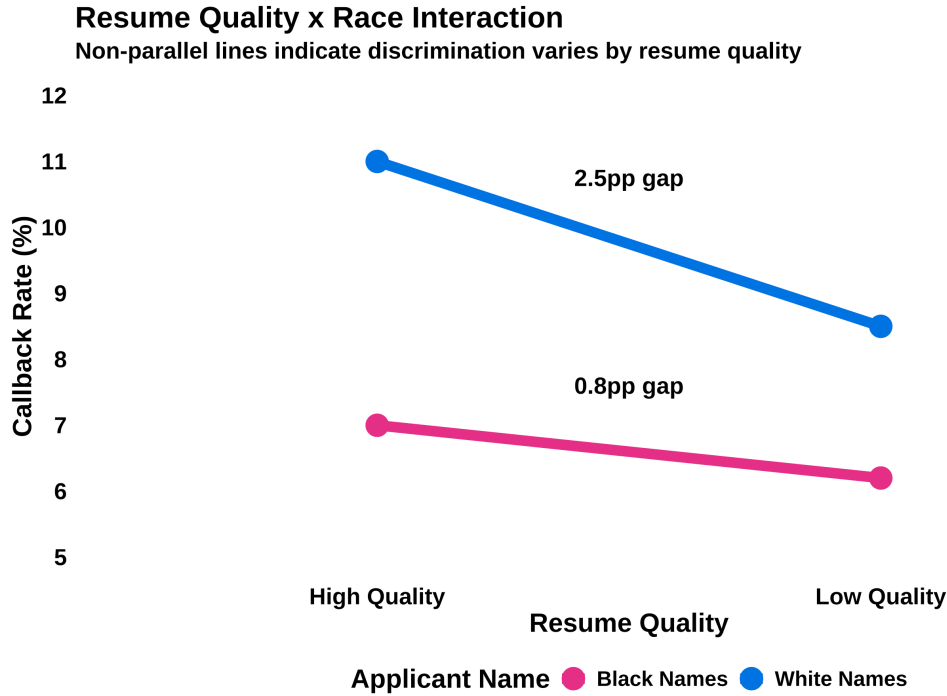
- Nearly 5,000 resumes sent
- Real job postings
- Chicago & Boston
- Field experiment



White names get 50% more callbacks. And resume quality helps White applicants more (2.3pp gain) than Black applicants (0.5pp).

Visualizing the Interaction

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 i Please use `linewidth` instead.



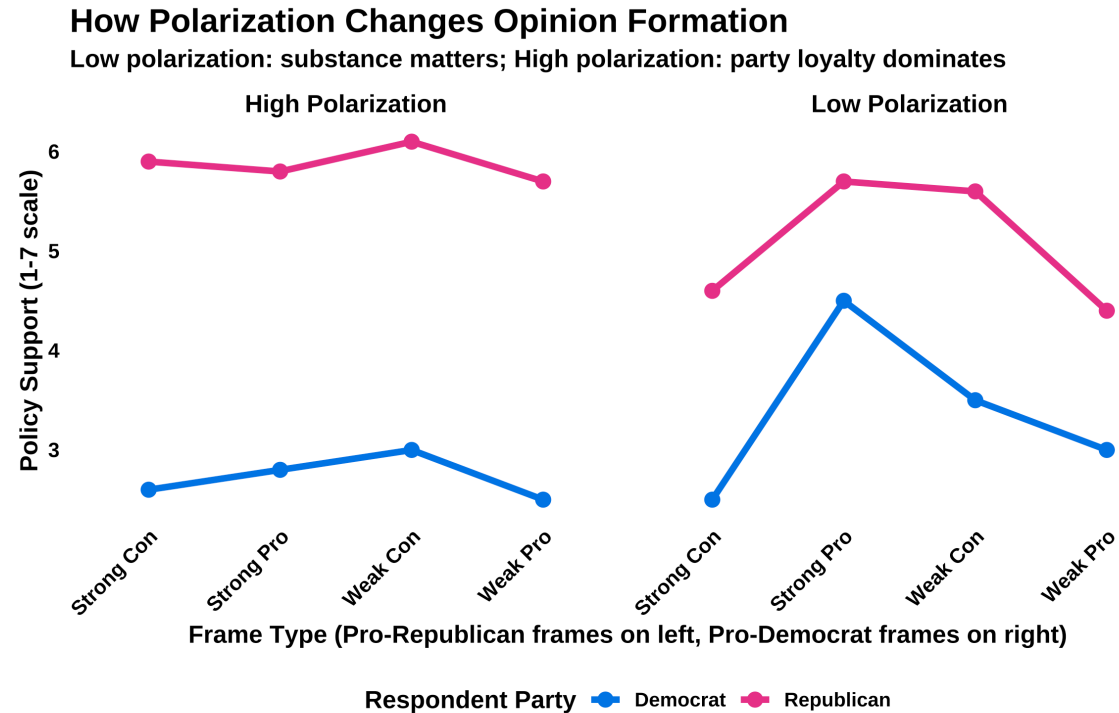
Polarization and Opinion Formation

Druckman, Peterson & Slothuus (2013) ran two experiments (drilling and the DREAM Act) with a three-factor design:

- Frame strength (strong vs. weak arguments)
- Party polarization (low vs. high elite polarization)
- Party cues (present vs. absent)

When polarization is low, strong frames win out over party cues — substance matters. When polarization is high, party cues dominate regardless of argument quality. Polarization makes people reason in partisan terms and feel more confident doing it.

Polarization Results



In the left panel, the lines cross — both parties respond to argument quality. In the right panel, the lines flatten and separate — party identity overrides everything.

Things That Go Wrong

The Hawthorne Effect

People change their behavior when they know they're being studied.

Origin: 1920s factory studies at the Hawthorne Works. Workers increased productivity regardless of the experimental manipulation — simply because they were being observed.

In political experiments: survey participation can increase political interest, and awareness of being studied can shift responses. This conflates the treatment effect with the observation effect.

Partial solutions: unobtrusive measures, natural field experiments, deception (with IRB approval).

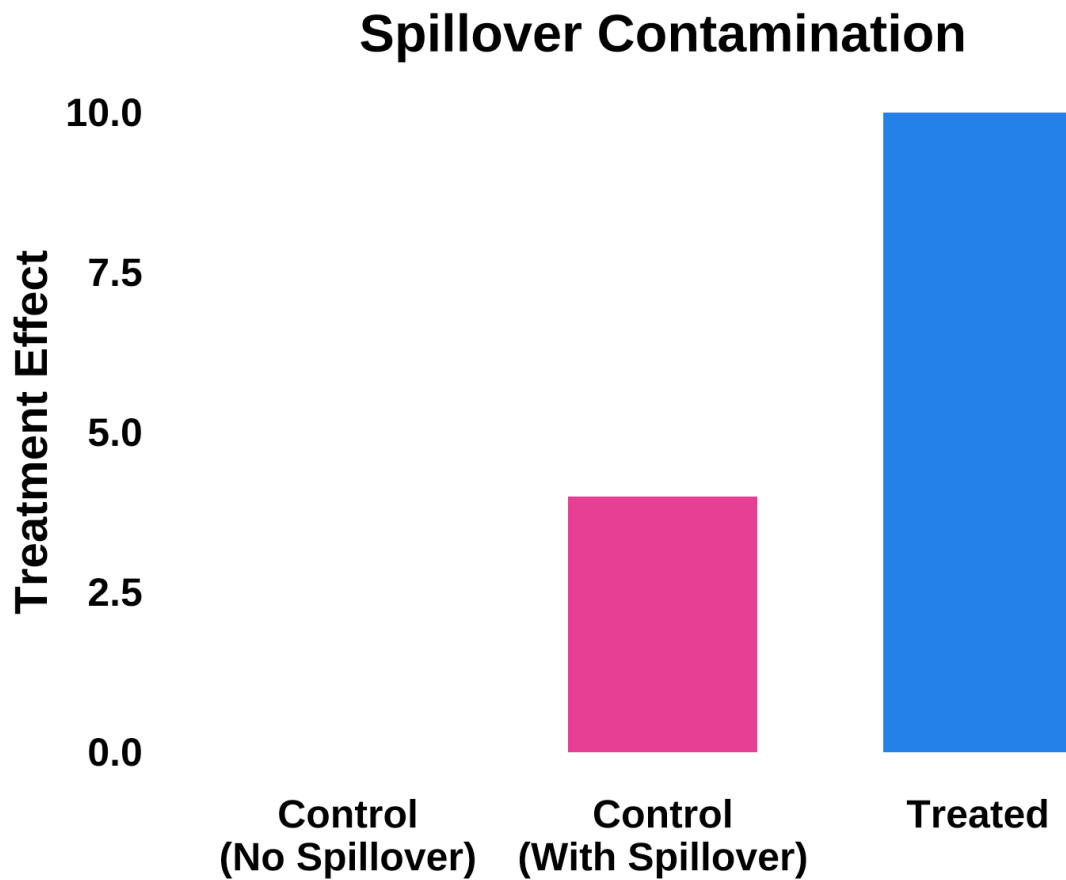
Spillover and SUTVA

SUTVA (Stable Unit Treatment Value Assumption) says your outcome depends only on *your* treatment assignment. It's violated when treatment “leaks” between units.

The assumption requires:

1. No interference between units
2. One version of treatment
3. Each person's outcome depends only on their own assignment

Violations are common: campaign ads reach neighbors, classroom interventions spread, social networks transmit information.

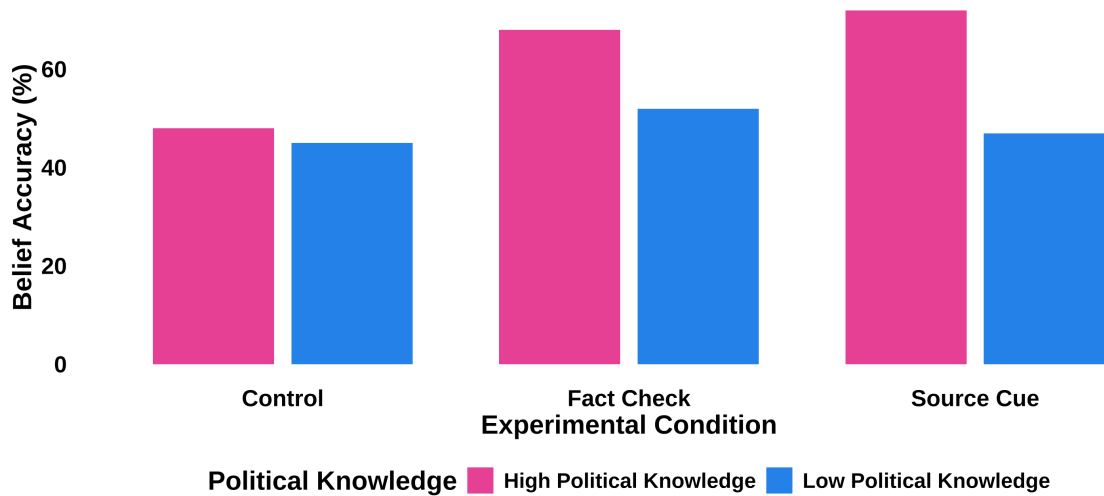


Heterogeneous Treatment Effects

Average effects can hide a lot. A treatment that “doesn’t work on average” might work very well for some people and backfire for others.

Heterogeneous Effects of Misinformation Corrections

Fact-checking helps low-knowledge citizens; source cues help high-knowledge citizens



The File Drawer and Pre-Analysis Plans

Franco, Malhotra & Simonovits (2014) looked at 221 TESS studies:

- 60% of strong results got published vs. 20% of null results
- 65% of null results were never even written up
- Strong results were 40pp more likely to be published

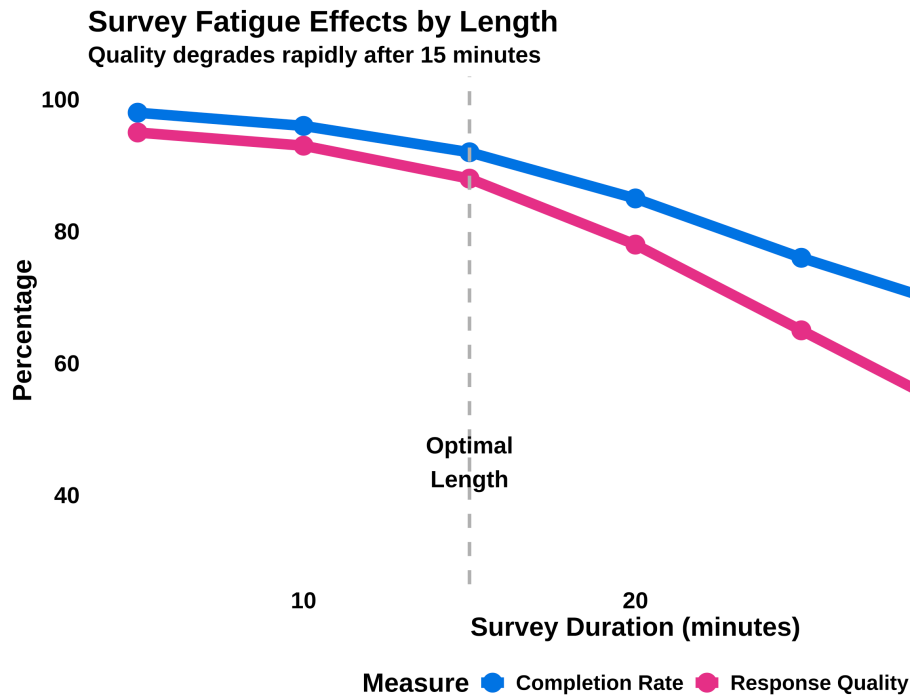
This is the file drawer problem: we only see what “worked.”

Pre-analysis plans (PAPs) are one fix. Before collecting data, you write down:

- Your hypotheses
- Your statistical tests
- Your outcome measures
- Your subgroup analyses

This locks in your analysis before you see the results, making it harder to p-hack and making null results more credible.

Survey Fatigue



Attention Checks and Data Quality

After randomization, the next threat is *who's even paying attention*. Online samples (MTurk, Prolific) include speeders, satisficers, and outright bots.

- **Attention checks / screeners:** simple items that catch people not really reading — e.g., “to show you’re paying attention, select ‘strongly disagree.’” Bots and click-through respondents fail them.
- They protect your data: inattentive responders add noise and can wash out a real effect.
- **Decide your exclusion rule in advance and pre-register it.** Dropping participants *after* you peek at the results is a back door to p-hacking.
- Keep checks simple and fair — a too-tricky check fails attentive people too.

Manipulation Checks

Verify that participants actually received and understood the treatment. Without this, a null result is ambiguous — maybe the treatment didn’t work, or maybe nobody noticed it.

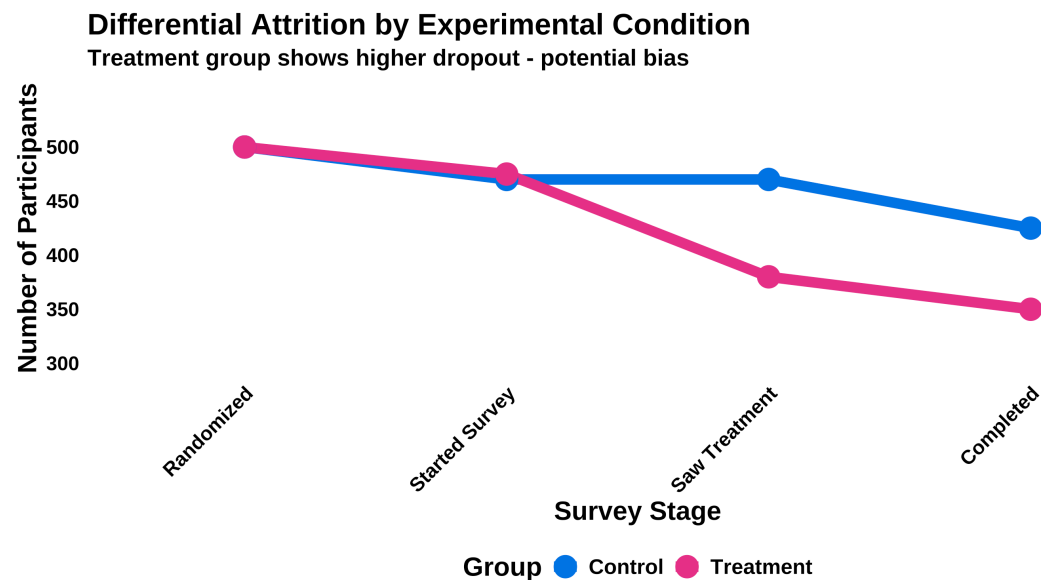
Place the manipulation check *after* your outcome measure. If you ask “did you notice the threat?” *before* measuring Y, you re-prime the treatment and risk manufacturing the very effect you are testing for.

Treatment Type	Example Treatment	Manipulation Check	Type
Information	Read article about trade	What was article about?	~75%
Framing	Issue framed as gain/loss	Was issue described positively?	~85%
Priming	Think about economy first	What did you think about?	~60%
Deception	False feedback on test	How did you perform?	~90%

Attrition and Non-Compliance

Two ways your experiment breaks down after randomization:

- **Attrition:** participants drop out (especially if the treatment is unpleasant)
- **Non-compliance:** participants don't follow their assignment



If dropout is differential — more people leave the treatment group — your treatment and control groups are no longer comparable. Intent-to-treat (ITT) analysis and bounds help, but prevention is better than cure.

Compliance: ITT vs. Treatment-on-Treated

Random assignment tells people what to do. It can't make them comply.

Get-out-the-vote: you assign 500 people to be canvassed, but only ~350 answer the door. Who is your “treatment group”?

Intent-to-Treat (ITT)

Compare everyone *assigned* to treatment vs. control — regardless of who complied.

Conservative, honest, and it keeps the randomization intact.

Treatment-on-Treated (ToT)

The effect among those who *actually* complied.

Bigger number — but compliers are a self-selected slice, no longer guaranteed comparable to controls.

Report **ITT** as your headline. It answers the question a campaign or policymaker actually faces: “if I *send* the canvassers, what happens?”

Differential Attrition: The Dangerous Kind

Some dropout is fine. *Differential* dropout is poison.

- If **15% leave both groups**, you lose statistical power but keep comparability.
- If more people leave the **treatment** group (say the treatment is unpleasant), the survivors in treatment differ from the survivors in control. Randomization is broken — and any gap in Y could be the dropout, not the treatment.

You can't fully fix this after the fact (bounds and ITT only help so much). **Prevent it by design:** keep the burden symmetric, don't make the treatment arm longer or more annoying, and track *who* leaves and from which arm.

Underpowered Studies Lie in Two Directions

Power is your chance of detecting a real effect. It rises with bigger effects, bigger samples, and less noisy outcomes.

An underpowered study isn't just “more likely to miss.” When a small study *does* hit significance, the effect is almost always **overstated** — you only cross the line when noise happens to break your way.

So a tiny experiment gives you the worst of both worlds: it misses real effects, and when it “finds” one, the size is inflated.

Estimate the sample you need *before* collecting data, for the smallest effect you'd care about. An honest null from a well-powered study beats an exciting result from an underpowered one.

Ethics

IRB

Institutional Review Boards exist because of historical abuses: Tuskegee, Nazi experiments, MK-Ultra. Since 1974, any research involving human subjects needs IRB approval.

The Belmont Report established three principles:

1. **Respect for persons:** autonomy and informed consent
2. **Beneficence:** maximize benefits, minimize harms
3. **Justice:** fair distribution of research burdens and benefits

Ethical Challenges in Political Experiments

Deception is sometimes necessary for validity — if participants know the real purpose, they change their behavior. But it must be justified scientifically, involve only minimal risk, and include thorough debriefing.

Example: telling participants they're evaluating "policies" when you're actually measuring prejudice.

Political science often studies sensitive topics:

- Racial attitudes
- Political violence
- Illegal behavior

These require extra care in design, consent procedures, and data protection.

Takeaways

Match the design to the question: vignettes for isolating single factors, conjoints for multidimensional preferences, games for behavioral measures with stakes, field experiments for real-world policy questions.

Within-subjects gives you more power but risks contamination across conditions. Between-subjects is cleaner but requires bigger samples. Factorial designs let you test interactions efficiently.

Implementation matters as much as design. Failed randomization, weak manipulations, and differential attrition can all undo a well-conceived study. Pre-register, check your manipulations, and report everything.

A well-designed experiment with 100 participants beats a poorly-designed study with 1,000.

Next Class

Transforming and creating variables — then causality: the three requirements for a causal claim, confounding variables, and spurious correlations.